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ing operation, the plunger is released at the push of a button and the molten solder then drawn up into the pump. Sometimes, it may take more than one attempt to clean the soldered point. You may add more solder to the joint and then desolder the whole quantity with the pump. Ensure that the components and PCBs are not damaged by overheating.

Using a desoldering braid or wick. The desoldering braid or solder wick is especially effective for desoldering a difficult joint. It is a simple and excellent alternative to a desoldering pump.

The desoldering braid, made from fine copper, draws molten solder up into the braid. To desolder using it, place a short length of the wick on the point to be desoldered. Place the hot bit of the soldering iron over the wick and press. The bit subsequently melts the solder and the molten lead is drawn up into the braid. Take extreme care to ensure no damage happens to the printed circuit track when the braid is pulled off the solder point.

Wave soldering

The wave soldering technique is used in the mass production of electronic circuits. It is an automated technique in which a large number of solder joints are made simultaneously. It reduces cost and increases the speed and reliability of production.

Wave soldering has three main requirements: solder, flux for wetting the soldering surface, and heat to activate the flux and melt the solder.

The PCB loaded onto a linear conveyor undergoes different operations, namely, fluxing, preheating and wave soldering. The conveyor is usually set at an angle of about 7° to 10° to the horizontal to enable excess molten solder to drain out from the board into the solder tray. For fluxing, a foamed standing wave or spraying is commonly used.

The next stage is preheating the PCB. Preheating is done to evaporate the flux solvent, activate the flux, and reduce thermal shock to the



Soldering wire

board when it touches the solder wave. Hot-air blowers or infrared heaters are commonly used.

Solder is heated in the solder bath

bed in the wave soldering stage. A solder wave is generated by continuously pumping molten solder in the reservoir. The wave thus formed is a standing wave with a cross-free, clean surface. The wave soldering machine can be set to adjust the shape and height of the wave. The assembled PCB, after fluxing and preheating, is passed over the solder wave. All solder points, which are in direct contact with the solder wave, establish perfect soldering.

Surface-mount soldering

The PCB is the heart of an electronics assembly. The quest for reliability and high efficiency in minimal space has led to the introduction of surface-mounted devices. Smaller components and higher packaging densities, with marked savings in size, weight and high-speed automated assembly, are the main advantages of surface-mount technology. This technology is widely used in pocket calculators, computers, communication field (especially in satellite communication), consumer electronics and automotive industry.

In this type of assembly, all the components are assembled on the same side. Solder paste is applied on the pads. The surface-mount components are then placed on the paste. Heat is applied to make the paste melt and flow, forming joints. Solder paste plays an important role in the assembly of surface-mount devices. The flux content of solder pastes makes the track surface clean, allowing perfect joints. ●

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